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Enhancing Semantic Scene Segmentation for Indoor Autonomous Systems Using Advanced Attention-Supported Improved UNet

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Abstract

This paper introduces EFFB7-UNet, an advanced semantic segmentation framework tailored for Indoor Autonomous Vision Systems (IAVSs) utilizing the U-Net architecture. The framework employs EfficientNetB4 as its encoder, significantly enhancing feature extraction. It integrates a spatial and channel Squeeze-and-Excitation (scSE) attention block, emphasizing critical areas and features to refine segmentation outcomes. Comprehensive evaluations using the NYUv2 Dataset and various augmented datasets were conducted. This study systematically compares EFFB7-UNet's performance with multiple U-Net encoders, including ResNet50, ResNet101, MobileNet V2, VGG16, VGG19, and EfficientNets B0-B6. The findings reveal that EFFB7-UNet not only surpasses these configurations in terms of accuracy but also highlights the effectiveness of the scSE attention block in achieving superior segmentation results. Without utilizing depth information, EFFB7-UNet achieves a 12% improvement in mean Intersection over Union (mIOU). This notable enhancement highlights EFFB7-UNet's adaptability across different domains, implying substantial progress in enhancing the effectiveness and reliability of Intelligent Autonomous Vision Systems (IAVS) technologies.

Keywords: Segmentation, UNet, Semantic Segmentation, Attention Mechanisms, Convolutional Neural Networks (CNN), EfficientNet

1 Introduction

Computer vision is a transformative branch of artificial intelligence that reshapes the way machines interpret visual information [1–8]. It has broad applications across various sectors such as healthcare imaging [9], autonomous vehicle technology [10], advanced robotics [11], security [12], and the construction industry [13]. A key process in computer vision is image segmentation [14], which simplifies the analysis and interpretation of images by dividing them into distinct sections. This technique is crucial for the precise identification and tracking of objects within an image, as it allows for detailed examination and classification.

Further exploration within image segmentation leads to semantic segmentation [15], which involves assigning class labels to each pixel, enabling machines to perceive depth in images similarly to human vision. This is especially important in indoor environments where there are many dynamic and constantly changing elements. Autonomous robots in human-centric environments often navigate these settings, where the background typically includes static features like floors and walls while the foreground is more dynamic with moving objects and people [15]. These robots must skillfully avoid obstacles to ensure safety.

For Indoor Autonomous Vision Systems (IAVSs), semantic segmentation is essential. It assigns a class label to every pixel, enhancing the understanding of and interaction with complex indoor scenes. Indoor environments, often being less structured and more intricate than outdoor ones, pose challenges to traditional scene parsing methods that depend on manual feature extraction, frequently yielding inferior results [16, 17]. The adoption of convolutional neural networks (CNNs) has significantly advanced semantic segmentation, improving the accuracy and efficiency with which IAVSs can process scenes [18, 19]. Moreover, incorporating depth data into image segmentation has been shown to enhance accuracy, as it provides vital contextual information about the spatial relationships of objects, aiding in their accurate recognition and categorization in complex indoor spaces [20, 21]. Researchers have employed two primary approaches to enhance the accuracy of segmentation models in deep learning. The first approach involves improving model precision by integrating depth data. However, the focus here is on the second approach, which centers on refining models that utilize only RGB data to achieve higher levels of accuracy without the use of depth images.

This paper concentrates on refining the UNet architecture through the introduction of an enhanced variant named EFFB7-UNet, which solely processes RGB data inputs, deferring the use of other data types. The key contributions of this research are outlined as follows:

1. **Integration of EfficientNetB7 Model:** The EFFB7-UNet incorporates the EfficientNetB7 as its encoder, substantially improving the model’s capability in feature extraction and thereby enhancing its ability to analyze complex image data effectively.
2. **Implementation of scSE Attention Mechanisms:** scSE attention mechanisms have been embedded within our network. These mechanisms focus on critical areas

and features within the images, fine-tuning the segmentation process and increasing accuracy, particularly around the edges and specific indoor image regions.

3. **Extensive Evaluation and Comparative Analysis:** Our model will be rigorously evaluated using the widely-used NYUv2 dataset, a common choice for studies involving depth data. This evaluation is designed to compare EFFB7-UNet’s performance with various UNet encoders, including ResNet50, ResNet101, ResNet18, MobileNet V2, VGG16, VGG19, and EfficientNets B0 to B6. The results demonstrate that EFFB7-UNet not only outperforms these configurations in accuracy but also underscores the scSE attention block’s role in enhancing segmentation outcomes.

This paper is structured as follows: Section II provides a comprehensive review of existing research in the field of semantic segmentation with a focus on indoor scenes. Section III discusses the proposed enhancements to the UNet architecture, detailing improvements such as the integration of the EfficientNetB7 encoder for superior feature extraction and the implementation of an advanced structured attention model for more precise segmentation of key regions. Section IV covers extensive experiments conducted on the NYU dataset, including an analysis of the results and a comparison of the performance of our EFFB7-UNet with other state-of-the-art approaches. Finally, Section V concludes the paper and outlines potential directions for future research.

2 Related Work and Proposed Method Overview

In the related work, Vijay et al. [22] simplified the NYU dataset’s comprehensive labels into 14 categories to improve the efficiency of semantic segmentation through the CNN-based SegNet framework. Daniel and colleagues [23] developed a semantic segmentation network using ResNet-18 and transfer learning tailored for indoor robotic navigation that effectively distinguishes between open floor spaces and walls. Additionally, Marchal et al. [24] utilized CNNs with the VGG16 architecture to segment indoor scenes, applying their method to the NYUv2 dataset.

Table 1 Assessment of semantic segmentation approaches on NYUv2 dataset

Model	Year	Pixel.acc (%)	m.IoU (%)	m.acc (%)
Segnet-Vijay et al. [22]	2015	-	-	50.5
VGG16-Marchal et al. [23]	2020	53.7	42.4	55.2
ResNet18-Daniel et al. [24]	2020	55.4	43.9	55.9
Proposed method	2024	62.33	55.19	65.84

According to Table I, when comparing these methods to our proposed approach, our model demonstrates superior accuracy, highlighting the effectiveness of our enhancements in semantic segmentation tasks. The EFFB7-UNet, an advanced model for semantic segmentation, is proposed, integrating EfficientNetB7 into its encoder to enhance feature extraction. The accuracy of the model is significantly enhanced by the utilization of EfficientNetB7. Additionally, the focus on segment boundaries is sharpened by the inclusion of a spatial and channel Squeeze-and-Excitation (scSE)

Attention Block, which markedly improves the precision of the segmentation outcomes. The EFFB7-UNet architecture is illustrated in Fig. 1 with detailed analyses of its constituent blocks provided in subsequent figures.

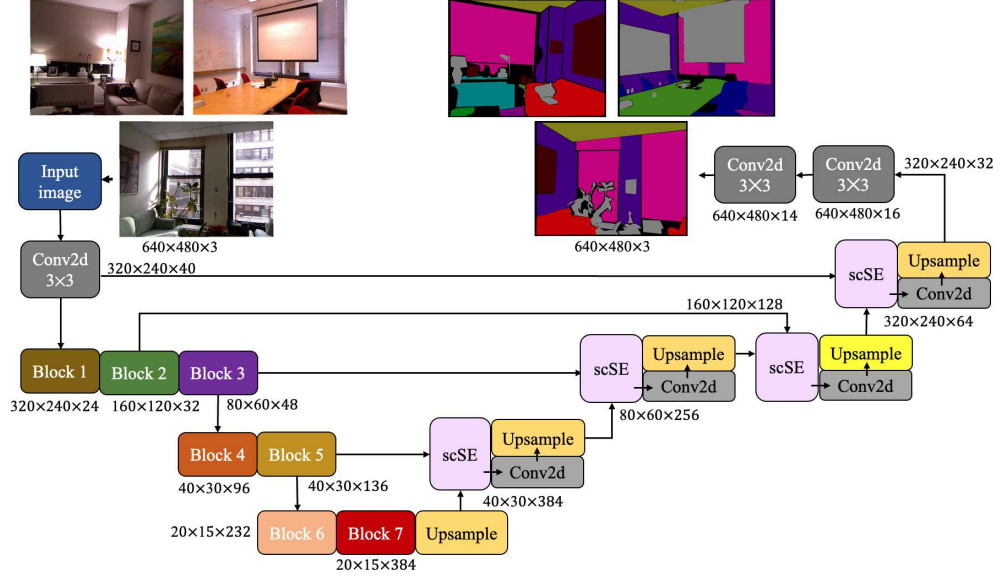


Fig. 1 The architecture of the proposed EFFB7-UNet framework for semantic segmentation with enhanced attention block (scSE).

3 Proposed UNet Architectures for Semantic Segmentation

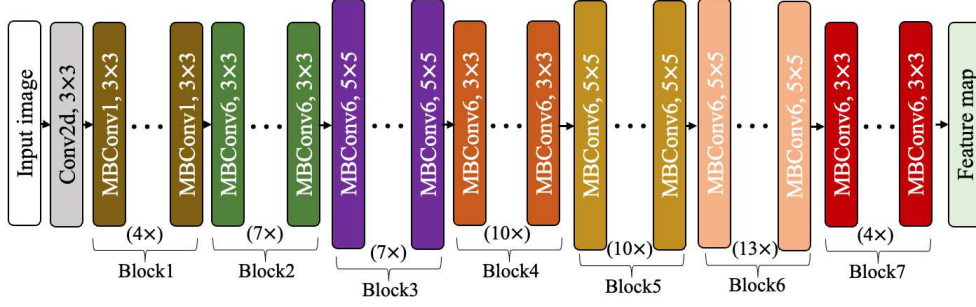
3.1 Enhancing the encoder structure of UNet with EfficientNetB7 for improved feature extraction.

An improved version of the UNet encoder architecture has been implemented, incorporating EfficientNetB7 from Google AI research for improved feature extraction capabilities. EfficientNet is a series of CNN architectures that are known for their incremental enhancements over previous models. This series includes eight models [25] ranging from B0 to B7, with each subsequent model displaying a growth in both parameters and accuracy. The comparison of metrics for the EfficientNet models from B0 to B7 is displayed in Table II.

In response to the accuracy requirements of the IAVS system, the encoder within the UNet architecture was upgraded to EfficientNetB7. Illustrated in Fig. 2, the encoder configuration is divided into eight phases, beginning with one Conv2d 3x3 stage, followed by seven additional blocks labeled 1 through 7. These blocks contain

Table 2 Comparison of metrics for the EfficientNet models from B0 to B7

Model	Input Size	Width coef.	Depth coef.	Parameter
EfficientNetB0	224x224	1.0	1.0	4057253
EfficientNetB1	240x240	1.0	1.1	6582914
EfficientNetB2	260x260	1.1	1.2	7777012
EfficientNetB3	300x300	1.2	1.4	10792746
EfficientNetB4	380x380	1.4	1.8	17684570
EfficientNetB5	456x456	1.6	2.2	28525810
EfficientNetB6	528x528	1.8	2.6	40973969
EfficientNetB7	600x600	2.0	3.1	64113049

**Fig. 2** Structure of EfficientNetB7 utilizing MBConv as fundamental components.

a total of 54 mobile inverted bottleneck convolutional (MBConv) stages, specifically four MBConv1s and 51 MBConv6s, as shown in Fig. 3.

The architecture of an MBConv includes an initial 1×1 convolution, a depthwise convolution followed by a sequence-and-excitation (SE) module, another 1×1 convolution to minimize dimensions, and concludes with a dropout layer. Batch normalization (BN) and Swish activation are applied after the initial 1×1 and depthwise convolutions. The subsequent 1×1 convolution focuses solely on BN operations. To improve the integration of feature data, a shortcut connection is utilized when the input and output feature matrices of the MBConv match. The SE module has played a crucial role in enhancing the accuracy of various tasks such as image classification, target detection, and image segmentation. The decoder consists of five up-sampling stages along with a sequence of convolution operations. The encoder's role in feature extraction and its subsequent restoration of the original image dimensions ($640 \times 480 \times 3$) for segmentation purposes are highlighted in Fig. 1. To refine feature specificity and mitigate noise, scSE attention blocks have been added to the skip connections, thus sharpening the precision of liver segmentation. Further details on the structure of the attention blocks will be discussed subsequently.

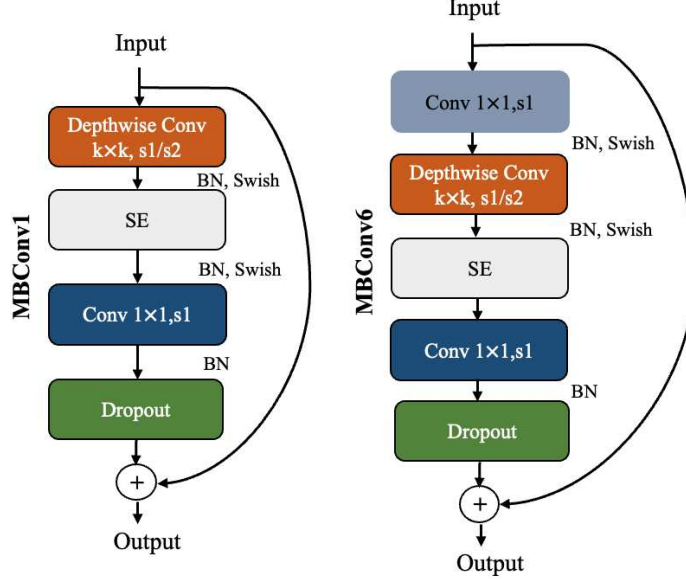


Fig. 3 MBConv1 and MBConv6: core components of the EfficientNet architecture.

3.2 Integration of Simultaneous Spatial and Channel Squeeze & Excitation (scSE) in UNet Architecture

To improve semantic segmentation in real-sense indoor robots, an attention module based on human attention mechanisms was introduced to enhance accuracy by focusing on information-rich areas rather than processing the whole image. A key development here is the integration of the scSE (Concurrent Spatial and Channel Squeeze & Excitation) attention block. This block enhances feature representation in Convolutional Neural Networks (CNNs) by simultaneously modulating spatial and channel data through two pathways: $\hat{C}_{sSE}$ (Channel Squeeze and Spatial Excitation) and $\hat{C}_{cSE}$ (Spatial Squeeze and Channel Excitation). These branches work together to recalibrate CNN outputs, focusing on key areas for better semantic segmentation. The intricate structure of the scSE block is illustrated in Fig. 4, demonstrating its role in directing network attention. In the scSE block, we have employed the Add algorithm to combine the outputs of the sSE (Channel Squeeze and Spatial Excitation) and cSE (Spatial Squeeze and Channel Excitation) pathways. This combination enhances the feature modulation capabilities of the block, effectively recalibrating both spatial and channel information within the network. The integration of these pathways is illustrated through the following (1):

$$\hat{C}_{scSE}(x, y, d) = \hat{C}_{sSE}(x, y, d) + \hat{C}_{cSE}(x, y, d) \quad (1)$$

where (x, y) is the spatial location at a specific depth d .

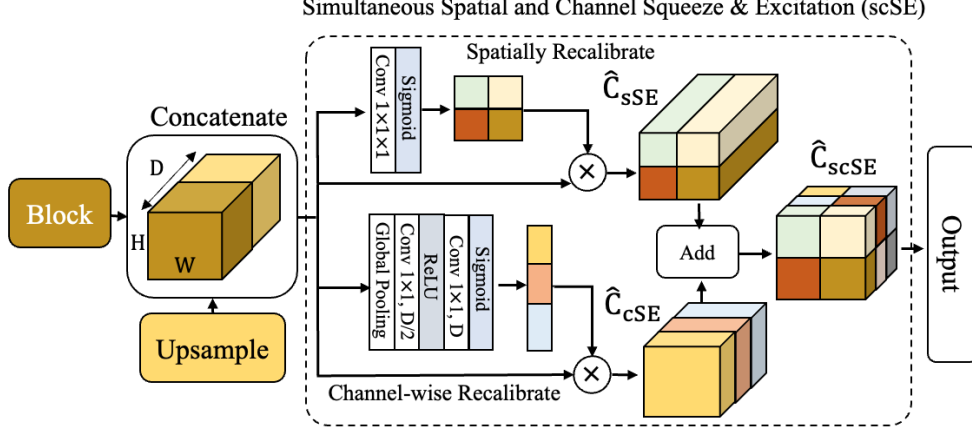


Fig. 4 Structure of the scSE Attention Block.

4 Experiments

4.1 Dataset and Experiment Setting

The EFB7-UNet model’s efficacy was evaluated using the indoor NYUv2 RGB dataset, which is segmented into training, validation, and testing sets accounting for approximately 76%, 14%, and 10% of the dataset respectively (Total 1449 images). This dataset adopts a classification system where pixels are allocated to one of 13 object classes with an additional category for non-labeled objects detailed in Fig. 5.

Developed on the PyTorch framework and leveraging an NVIDIA Tesla T4 15GB GPU, the EFB7-UNet underwent training with data augmentation techniques such as flipping, grid distortion, and the introduction of random noise and lighting adjustments to enrich the training dataset. For validation, horizontal flips and grid distortion were selectively applied to broaden the diversity of the data. The images for training and validation were uniformly resized to a resolution of 640×480 pixels. The training was executed throughout 20 epochs, maintaining a batch size of 1 and a starting learning rate of 0.00015 to optimize model performance.

4.2 Evaluation Metrics and Results

Evaluation Metrics: Four commonly used metrics in evaluations of semantic segmentation and scene parsing were employed to analyze the results as described in [26]. These measures are Pixel Accuracy (Pixel.acc), Mean Accuracy (m.acc), Mean Intersection Over Union (m.IoU), and Frequency Weighted Intersection Over Union (F.w.IoU) detailed as follows:

$$Pixel.acc = \frac{\sum_i x_{ii}}{\sum_i T_i} \quad (2)$$

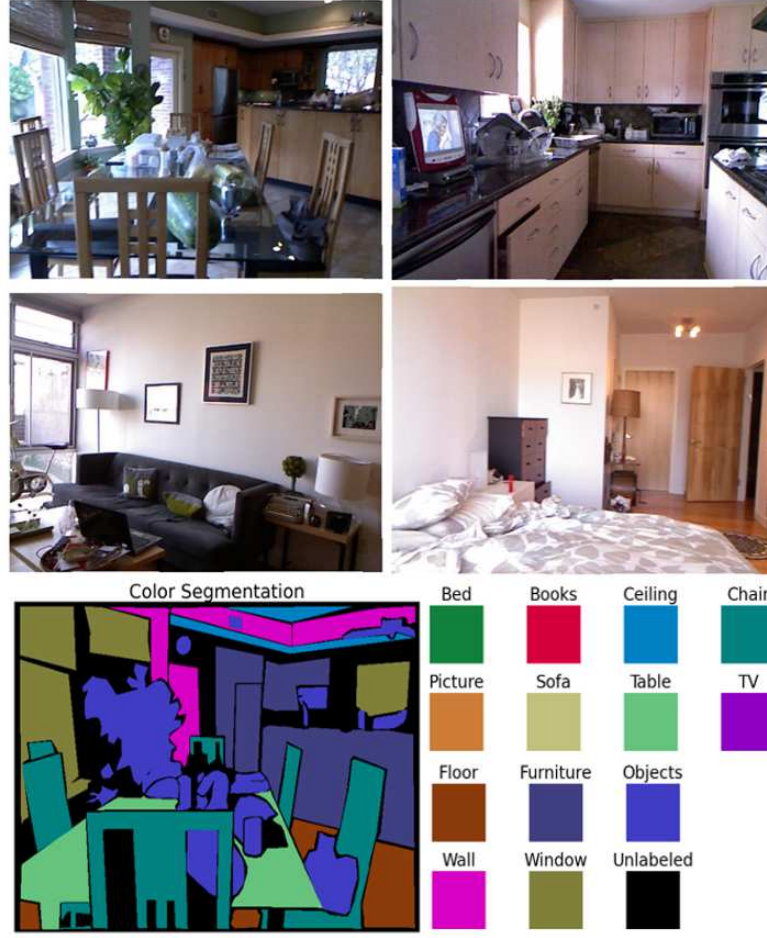


Fig. 5 The NYUv2 dataset features RGB images along with a truth mask where each pixel is tagged with unique colors that match the specified color scheme.

$$m.acc = \frac{1}{n_C} \sum_i \frac{x_{ii}}{T_i} \quad (3)$$

$$m.IoU = \frac{1}{n_C} \sum_i \frac{x_{ii}}{T_i + \sum_j x_{ji} - x_{ii}} \quad (4)$$

$$F.w.IoU = \left(\sum_k t_k \right)^{-1} \sum_i \frac{T_i \cdot x_{ii}}{T_i + \sum_j x_{ji} - x_{ii}} \quad (5)$$

where x_{ij} is the number of pixels from class i that are classified as class j with n_C representing the total number of unique classes and $T_i = \sum_j x_{ij}$ representing the sum of pixels that are a part of class i .

In the training process, the Cross Entropy (CEL) loss function is utilized. All models compared in this study employ this particular loss function. However, after

identifying the best-performing model, we also conducted further testing to assess the results when alternative loss functions are applied, such as Focal loss (FL) and Dice Loss (DL). The formulas for these loss functions are presented below.

$$CEL = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^n y_{ij}^T \log y_{ij}^P \quad (6)$$

$$DL = \frac{1}{n} \sum_{j=1}^n \frac{1}{N} \left(1 - 2 \frac{\sum_{i=1}^N y_{ij}^T y_{ij}^P}{\sum_{i=1}^N (y_{ij}^T)^2 + \sum_{i=1}^N (y_{ij}^P)^2} \right) \quad (7)$$

$$FL = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^n (1 - y_{ij}^P)^\gamma y_{ij}^T \log y_{ij}^P \quad (8)$$

where N represents the total number of input pixels, n denotes the number of semantic categories, y_{ij}^T indicates the true label for the segmentation task, and y_{ij}^P represents the predicted label in the segmentation results. Additionally, γ is the exponent factor used within the Focal Loss function.

Table 3 Assessment of semantic segmentation approaches on NYUv2 dataset

Model	Pixel.acc (%)	m.IoU (%)	m.acc (%)	F.w.IoU (%)
ResNet50	48.66	35.64	50.52	38.22
ResNet101	54.21	42.20	55.25	44.76
MobileNet-v2	54.32	43.33	54.76	45.52
VGG16	52.74	41.44	54.23	43.41
VGG19	49.70	38.50	51.30	40.50
EfficientNetB0	58.83	47.85	59.87	49.68
EfficientNetB1	59.11	48.89	60.54	50.73
EfficientNetB2	59.90	49.12	61.33	51.23
EfficientNetB3	61.10	50.25	62.61	51.94
EfficientNetB4	61.17	50.84	62.98	51.97
EfficientNetB5	61.23	51.22	63.22	52.05
EfficientNetB6	61.42	52.55	63.87	52.15
EfficientNetB7	61.75	53.23	64.20	52.24

Results and Discussion: In the study presented in Table III, a comprehensive analysis has been performed to evaluate the accuracy of different backbones used in semantic segmentation on the NYUv2 dataset. The backbones evaluated include a range of commonly employed architectures in semantic segmentation tasks such as ResNet50, ResNet101, MobileNetV2, VGG16, VGG19, and a suite of EfficientNet backbones from B0 to B7, all incorporated within a UNet framework. As detailed in Table III, the proposed EfficientNet-B7 backbone, when integrated with the UNet architecture, emerges as the superior performer. It showcases a notable enhancement in accuracy over the other models. The findings underscore the efficacy of the UNet model equipped with the EfficientNet-B7 backbone as it not only surpasses the accuracy metrics of other widely recognized methods but also does so with a relatively moderate parameter count. This balance of high accuracy and efficiency in parameter usage sets the EfficientNet-B7 apart as a benchmark in the field of semantic segmentation.

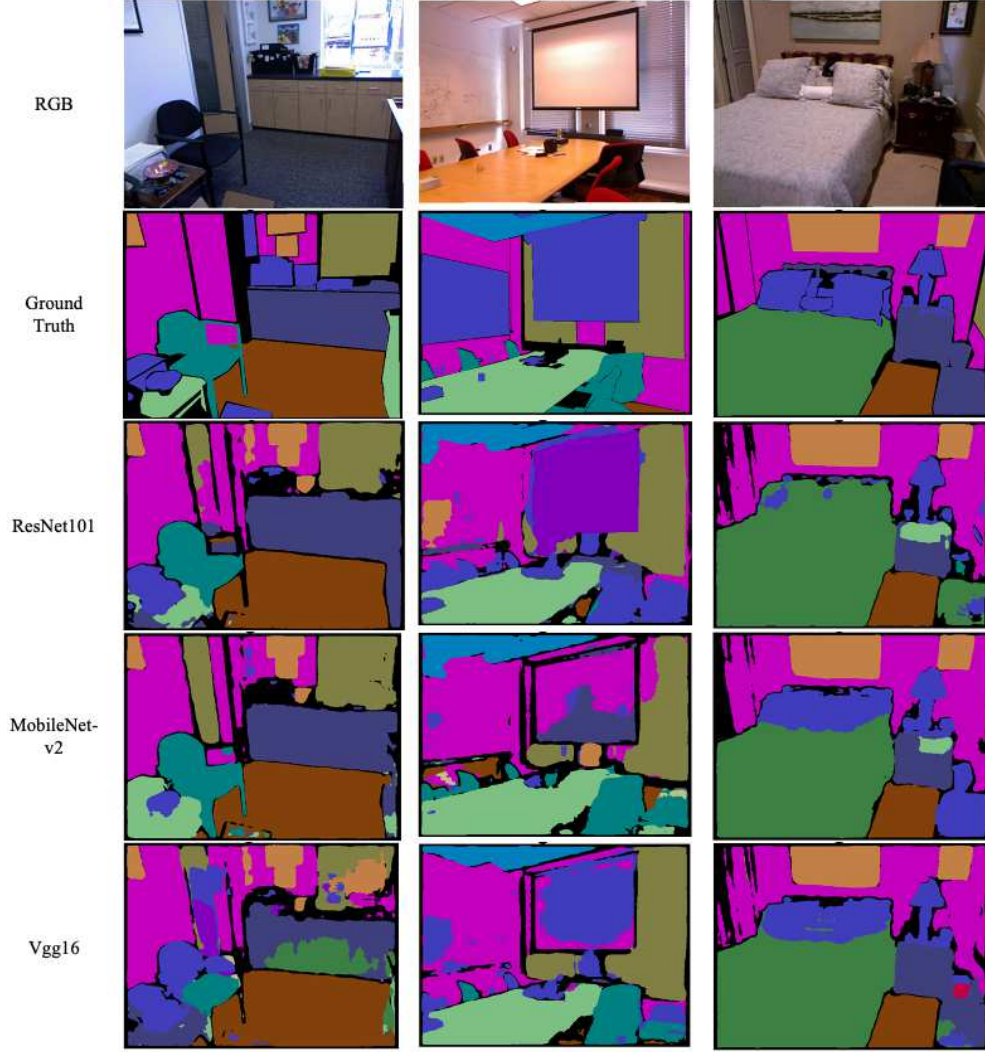


Fig. 6 The image exhibits a side-by-side comparison of RGB images with their respective ground truth masks from the NYUv2 dataset. It also showcases the segmentation results produced by processing the images with a range of distinct encoder backbones.

During the training process, the Cross-Entropy Loss (CEL) function was employed across all methods. Fig. 6, 7 provides an illustrative display of the segmentation outputs from all the approaches in the study. This figure offers a detailed and clear view, allowing for an easy comparison of the segmentation effectiveness. It is readily observable that the proposed method yields highly accurate results.

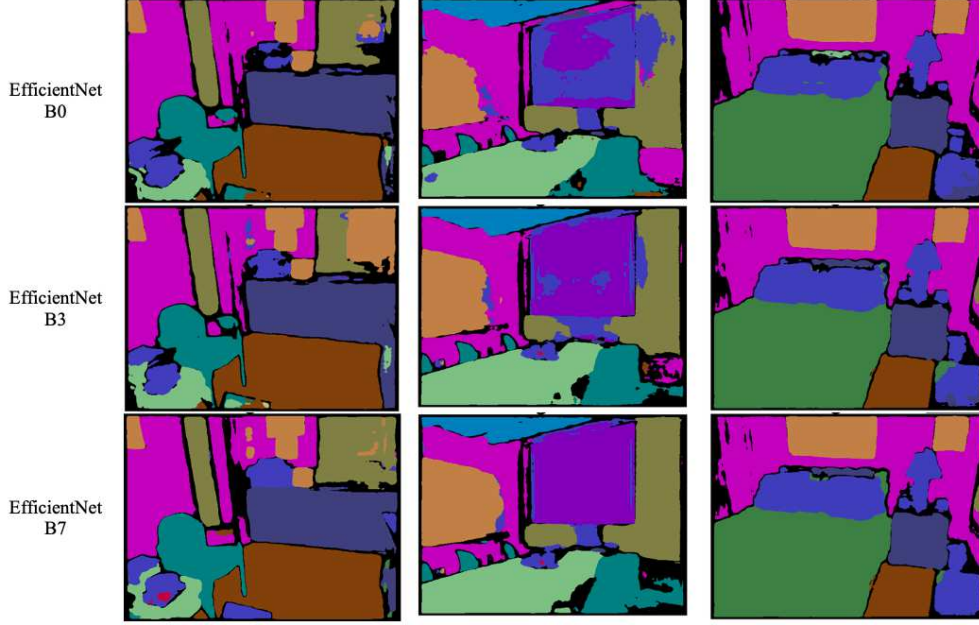


Fig. 7 The image exhibits a side-by-side comparison of RGB images with their respective ground truth masks from the NYUv2 dataset. It also showcases the segmentation results produced by processing the images with a range of distinct encoder backbones.

Table 4 Comparison of Segmentation Results Using Various Common Loss Functions with and without scSE Attention on the EfficientNetB7 Encoder UNet Backbone.

Model	Pixel.acc (%)	m.IoU (%)	m.acc (%)	F.w.IoU (%)
No attention + CEL	61.75	53.23	64.20	52.24
scSE attention + CEL	62.33	55.19	65.84	53.33
scSE attention + FL	61.22	54.73	65.52	51.88
scSE attention + DL	61.32	54.28	65.63	52.34

Our study in Table IV focuses on the impact of different loss functions on the EfficientNetB7-UNet model for semantic segmentation on the NYUv2 dataset. Implementing the scSE attention mechanism with Cross-Entropy Loss (CEL) provided the best overall improvement in accuracy, outperforming the standard CEL as well as Focal Loss (FL) and Dice Loss (DL) combinations. While the scSE attention with FL and DL also showed promising results, the scSE+CEL configuration yielded the highest performance increases, making it the preferred choice for enhancing semantic segmentation with this model.

The visual illustration in Fig. 8 effectively showcases the comparative segmentation results of the EfficientNetB7-UNet model with and without the incorporation of the spatial and channel squeeze and excitation (scSE) attention blocks. A discernible difference is evident in the accuracy of the boundary regions classified between different objects. The incorporation of scSE attention blocks significantly enhances the

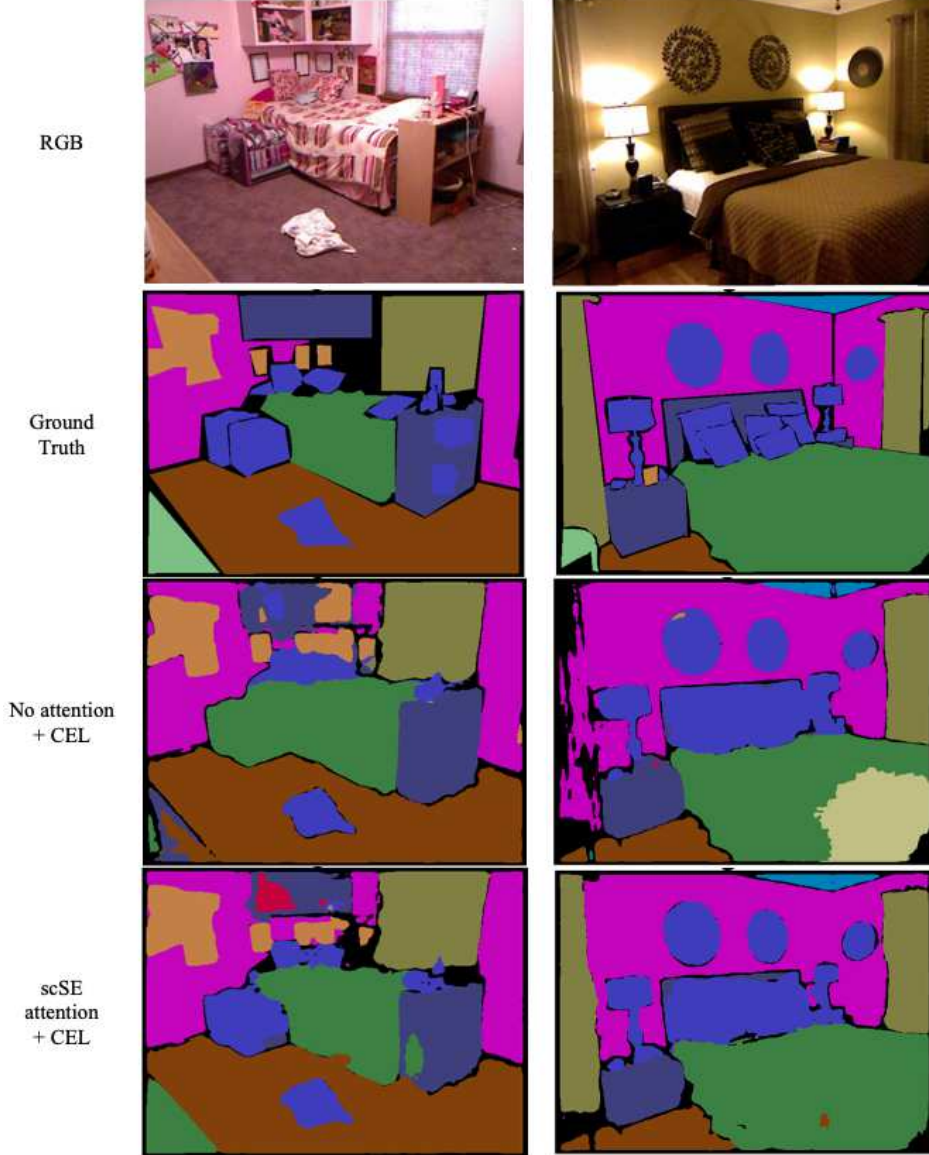


Fig. 8 Comparative analysis of EFFB7-UNET segmentation outcomes with and without scSE Attention on NYUv2 Dataset.

model's fidelity in segmenting adjacent yet distinct regions. This is particularly noticeable when the segmented pixels are juxtaposed with the ground truth, where the model employing scSE attention exhibits higher precision, suggesting a marked improvement in capturing the nuanced contours of the segmented objects.

In Fig. 9, the graph detailing 20 epochs of model training and validation presents insightful trends regarding the performance of the EFFB7-UNET architecture under

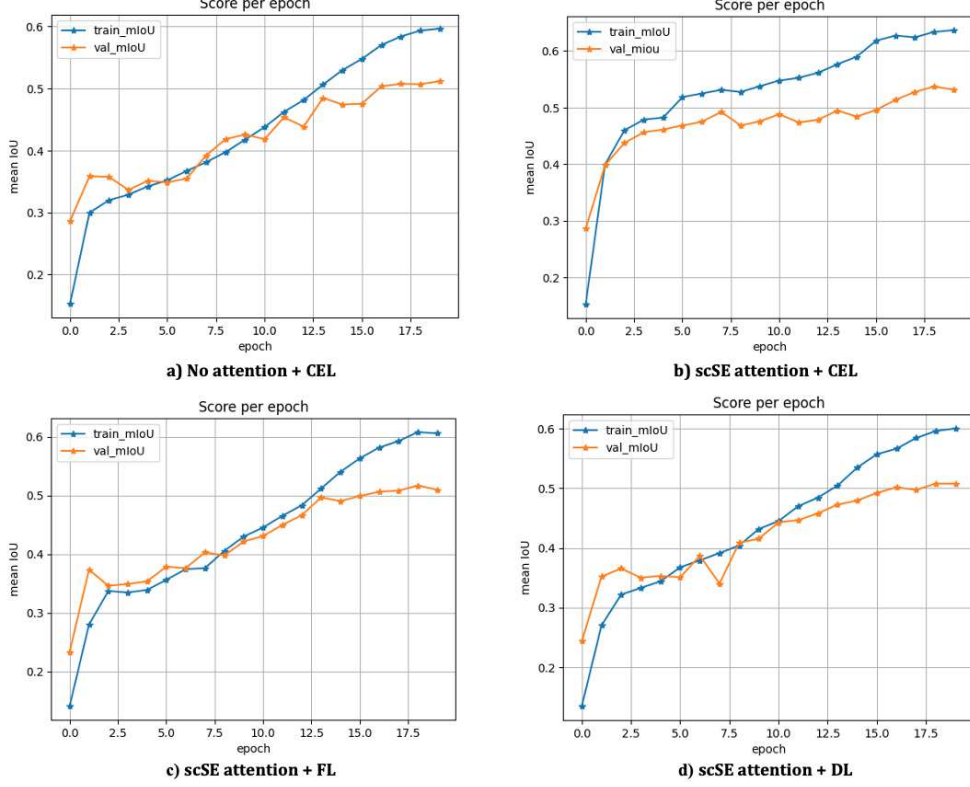


Fig. 9 Comparative analysis of training mIoU, validation mIoU over 20 epochs for three variants: EFFB7-UNET without attention block + CEL, with scSE attention block + CEL, with scSE attention block + FL, and with scSE attention block + DL on the NYUv2 Dataset.

different configurations. Notably, the EFFB7-UNET model utilizing the Cross-Entropy Loss (CEL) function consistently achieves the highest Mean Intersection over Union (mIoU) scores when compared to the other three methods. This indicates a strong convergence and generalization capability of the model with the CEL function underpinning its effectiveness in semantic segmentation tasks on the NYUv2 dataset.

5 Conclusion and Future Scope

This paper proposes EFFB7-UNet, a novel semantic segmentation framework tailored for Indoor Autonomous Vision Systems (IAVSs) using the U-Net architecture. By employing EfficientNetB7 as its encoder and integrating a spatial and channel Squeeze-and-Excitation (scSE) attention block, EFFB7-UNet enhances feature extraction and refines segmentation outcomes. Further exploration of alternative loss functions beyond Cross-Entropy Loss (CEL) could provide valuable insights for improving semantic segmentation tasks. The proposed approach has been validated through comprehensive experimentation, including comparative visualizations of output results among various backbone models. By examining the utilization and omission

of attention mechanisms, as well as the application of different loss functions such as Focal Loss (FL) and Dice Loss (DL), the effectiveness of EFFB7-UNet has been demonstrated. These experiments provide empirical evidence supporting the superior performance of EFFB7-UNet in semantic segmentation tasks, contributing to advancements in the field of Intelligent Autonomous Vision Systems (IAVS).

Future work will focus on applying the EFFB7-UNet framework to robotic models for real-world deployment. By integrating EFFB7-UNet into robotic vision systems, we aim to enhance their semantic segmentation capabilities, enabling more accurate and reliable perception in dynamic environments. Additionally, ongoing research will concentrate on further refining the framework to improve segmentation accuracy, particularly in challenging scenarios such as low-light conditions or cluttered scenes. Moreover, exploring techniques to optimize computational efficiency without compromising performance will be a key area of investigation, ensuring that the model can operate effectively in resource-constrained robotic platforms.

Author Contributions HNT conceived the research idea, developed the algorithms, and conducted the simulation experiments. All authors contributed to the analysis of the results. NVN, NQPL, and NNNN prepared the initial draft of the manuscript. TANL and VDN oversaw the research process and made revisions to the manuscript. All authors reviewed and approved the final version of the manuscript.

Data Availability This research employed the NYUv2 indoor RGB dataset, linked to the paper by Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus, titled "Indoor Segmentation and Support Inference from RGBD Images." The dataset is publicly available at the following URL: [NYU Depth V2 Dataset](https://cs.nyu.edu/~silberman/datasets/nyu_depth_v2.html)

Code Availability The code will be made available upon request.

Declarations

Ethics Approval Not applicable. (This study does not involve human participants, their data, or biological material).

Consent to Participate Not applicable (This article does not involve human participants).

Consent for Publication Not applicable (This article does not involve human participants).

Conflicts of interest The authors declare neither conflict of interest, nor competing interests.

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